Operating Systems · 21CSC202J

Unit 1 · Session 10 · SLO 2 — Solved Worksheet

Name: R L KRISHE YADAV R L KRISHE YADAV | **Reg. No:** RA2511030010473

Unit â€“ 1 : Session â€“ 10 : SLO â€“ 2 | User Mode to Kernel Mode Transition â€” Scrambled Sentences

I. Sentences in Correct Order (System Call Transition)

1. A user program runs in USER MODE with limited privileges.
2. When the program needs an OS service (I/O, file access), it invokes a SYSTEM CALL.
3. The system call triggers a TRAP (software interrupt) that switches the CPU to KERNEL MODE.
4. The kernel validates the request and executes the corresponding service routine.
5. The result is returned and the CPU switches back to USER MODE.
6. The user program resumes execution with the system call's return value.

II. Key Point

• Mode bit in the CPU distinguishes kernel mode (0) from user mode (1); the trap instruction is the
• controlled gateway that flips the bit, keeping user programs from executing privileged instructions.